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PROJECT TECHNICAL REPORT
WEB TECHNOLOGY TECHNICAL ASSESSMENT 2026

Project Title: TUGENDE — Online Bus Ticket Booking System

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Note: page-reference numbers will populate automatically once figure captions are refreshed in Word (References > Update Table).

List of Abbreviations

Abbreviation	Meaning
UI	User Interface
UX	User Experience
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
ERD	Entity Relationship Diagram
DB	Database
API	Application Programming Interface
RWF	Rwandan Franc
MoMo	Mobile Money
SMS	Short Message Service
ID	Identity Document
FAQ	Frequently Asked Questions
VIP	Very Important Person (premium bus class)

Chapter 1. Introduction

1.1. Historical Background of the Case Study

Inter-city bus travel is the backbone of public transport in Rwanda, moving passengers daily between Kigali and provincial towns such as Musanze, Huye, Rubavu and Nyamata. Historically, tickets for these routes have been sold almost exclusively at physical terminal counters — most prominently Nyabugogo Terminal in Kigali and the Musanze Bus Park — where travelers queue in person, pay in cash, and receive a paper ticket immediately before departure.

As smartphone and mobile-money adoption has grown rapidly across Rwanda, bus operators have come under increasing pressure to modernize how they publish schedules and sell tickets. This case study — presented under the working brand "Tugende" / "Tugende Express" — began as a Design Thinking exercise for the WFEN 100 Orientation Program in Entrepreneurship module (Group 5), in which the team empathized with real commuters before defining and prototyping a redesigned booking experience. The web front end analyzed in this report is the resulting implementation of that process.

1.2. Problem Statement

The team's Empathize-mode interviews with daily commuters, students and elderly travelers surfaced a consistent set of pain points with existing online bus-booking platforms:

- Slow page loads and frequent crashes during peak booking hours — one commuter reported: "The site freezes right when I try to pay — I've missed my bus waiting for it to load."
- No visual seat map, so seat numbers are effectively picked blind rather than chosen the way a flight-booking app allows.
- A confusing, multi-step payment process with a high failure rate.
- Difficult cancellations and refunds once a booking is made.
- Interfaces that are not friendly to elderly or first-time users, who often have to ask a family member to complete a booking for them.

Distilled into a single Define-mode problem statement, the team framed the challenge as: bus commuters need a fast, reliable and simple way to book and cancel tickets online — especially on mobile — because current platforms are slow, confusing and unreliable, causing

missed trips and lost trust. The web pages analyzed in this report (schedules, reservation form, login, support) are the team's response to that statement.

1.3. Objectives of the Project

General objective: to design and develop a web-based bus ticket reservation system for Tugende Express that allows passengers to view schedules and reserve tickets remotely, while giving staff a separate administrative entry point.

In the Ideate mode, the team generated and compared three candidate solutions — a Mobile-First Booking App with a real-time seat map and 3-step checkout, an SMS/Chat Booking Bot for feature-phone users, and a Multi-Operator Aggregator comparing routes and prices across operators — and selected the mobile-first booking concept as the direction to prototype. That choice shapes the specific objectives below:

- Provide a public dashboard that presents route schedules, fares, bus types and terminal information at a glance, addressing the "visual clarity" and "speed & reliability" pillars identified in Define mode.
- Provide a reservation form through which a passenger can supply personal details, choose a route, date, time, seat preference and payment method, as a first step toward the mobile-first, low-friction checkout envisioned in Ideate mode.
- Provide separate, clearly distinguished login entry points for passengers ("User Login") and staff ("Admin Login").
- Provide a Help & Support / FAQ page covering common questions, an inquiry form and emergency contact numbers, directly answering the elderly/first-time-user friction raised in Empathize mode.
- Lay out a clear, extensible interface structure that a future back end (database, authentication, a real-time visual seat map and payment processing) can be attached to.

1.4. Expected Outcomes

By the end of the project, a functioning static web application is expected, comprising: a landing dashboard that frames the schedule page and the login panel side by side; a bus-information page listing weekly departures, fares and amenities; a reservation form that captures all details needed to issue a ticket; distinct user and admin login pages; and a support page combining FAQs, an inquiry form and contact channels. Together these pages

demonstrate the front-end user journey of an online bus-ticketing platform, ready to be connected to a persistence layer in a later iteration.

The team's own Prototype-mode success measures — set out for the fuller mobile-first concept and validated on a separate live prototype at tugende-e.onrender.com — target an average end-to-end booking time under two minutes, real-time seat availability with no blind picks, instant e-ticket delivery by SMS and email, and a 1-tap cancellation/refund flow. The implementation reviewed in this report (Chapter 3) currently delivers the schedule browsing, reservation-form and support pieces of that vision; the real-time seat map and automated e-ticket delivery remain future work, discussed in Chapter 4.

1.5. Project Rationale (Importance to the Community)

A working online booking front end reduces the time passengers spend queuing at terminals, gives them visibility into fares and schedules before they travel, and opens the door to cashless payment via mobile money. For the operator, centralizing terminal information, policies and support contacts on one site reduces the volume of routine phone enquiries and presents a more professional public image. For the specific commuters interviewed during Empathize mode — daily riders anxious about payment failures, students wanting to compare seats the way they would on a flight app, and elderly travelers who currently need help from family to book at all — a faster, clearer, mobile-first booking flow directly addresses a friction point in their daily lives. More broadly, the project contributes to the wider digitization of Rwanda's transport sector, an area identified as a priority for improving mobility and reducing informal-sector friction in public transport.

Chapter 2. System Analysis and Design

2.1. System Analysis

2.1.1 Intended Users of the Project

- Passengers / general public — browse schedules and terminal information, log in, and submit ticket reservations.
- Bus company staff / administrators — log in through a dedicated admin panel to (in the full system) manage routes, schedules and bookings.
- Prospective and existing customers seeking support — use the FAQ and inquiry form for questions about refunds, lost items, or policy.

2.1.2 Intended System Partners

- Bus terminals: Nyabugogo Terminal (Kigali, main station), Musanze Bus Park, Huye Main Station, Rubavu Bus Terminal and Nyamata Bus Stop.
- Mobile money and banking providers, to support the "Mobile Money (MoMo)" and "Bank transfer" payment options offered on the reservation form.
- An SMS gateway provider, to deliver the booking-confirmation messages referenced on the reservation and information pages.
- Rwanda National ID / passport verification, since travelers are asked to carry an ID for verification when boarding.

2.2. System Design

1. High-Level Architecture

The current implementation is a static, client-side web application built with plain HTML, CSS and a small amount of JavaScript (for the live clock on the dashboard). `index.html` acts as a dashboard shell: it lays out a two-column frame in which `home.html` (schedules and terminal information) is embedded on the left via an `iframe`, and the login panel — toggling between `user_login.html` and `admin_login.html` — is embedded on the right. From the schedule page, passengers can navigate to `reserve_ticket.html` to submit a booking, or to `faqs.html` for support.

The diagram below also sketches the server tier that a production deployment would need — an application/API server, a relational or document database, and a gateway for SMS/MoMo integration — none of which is implemented yet; the current milestone covers the client tier only.

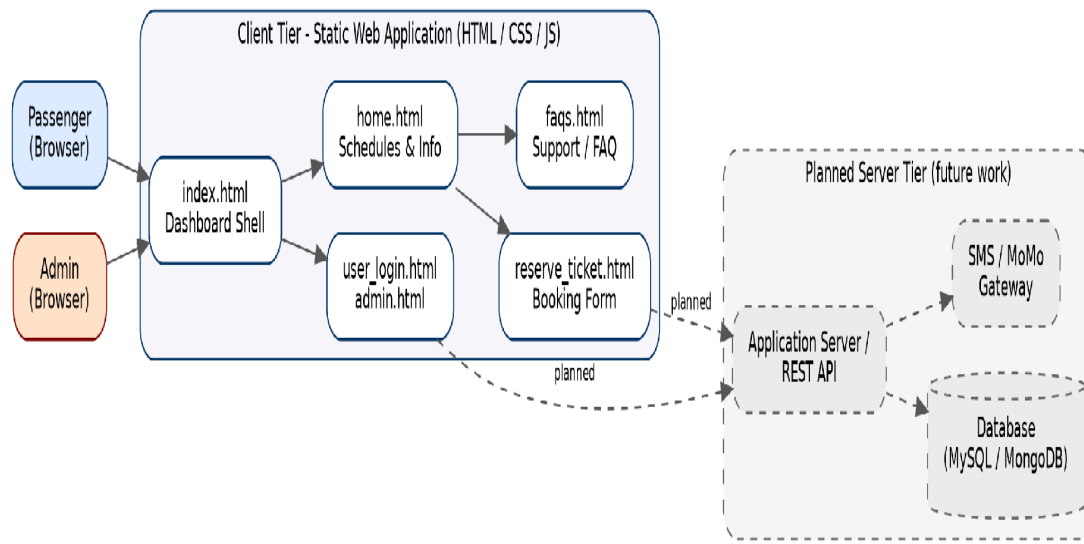


Figure 1: High-level system architecture of the Tugende platform

2. UI Design (Sketch of the UI Module) and Flowcharts

The interface follows a consistent visual language across all pages: a dark-blue (#003366) header/footer band, an orange (#FF9900) accent band for sub-headers and calls to action, and rounded form controls defined in styles.css. The typical passenger journey — from arriving at the dashboard to receiving a booking confirmation — is summarized in the flowchart below.

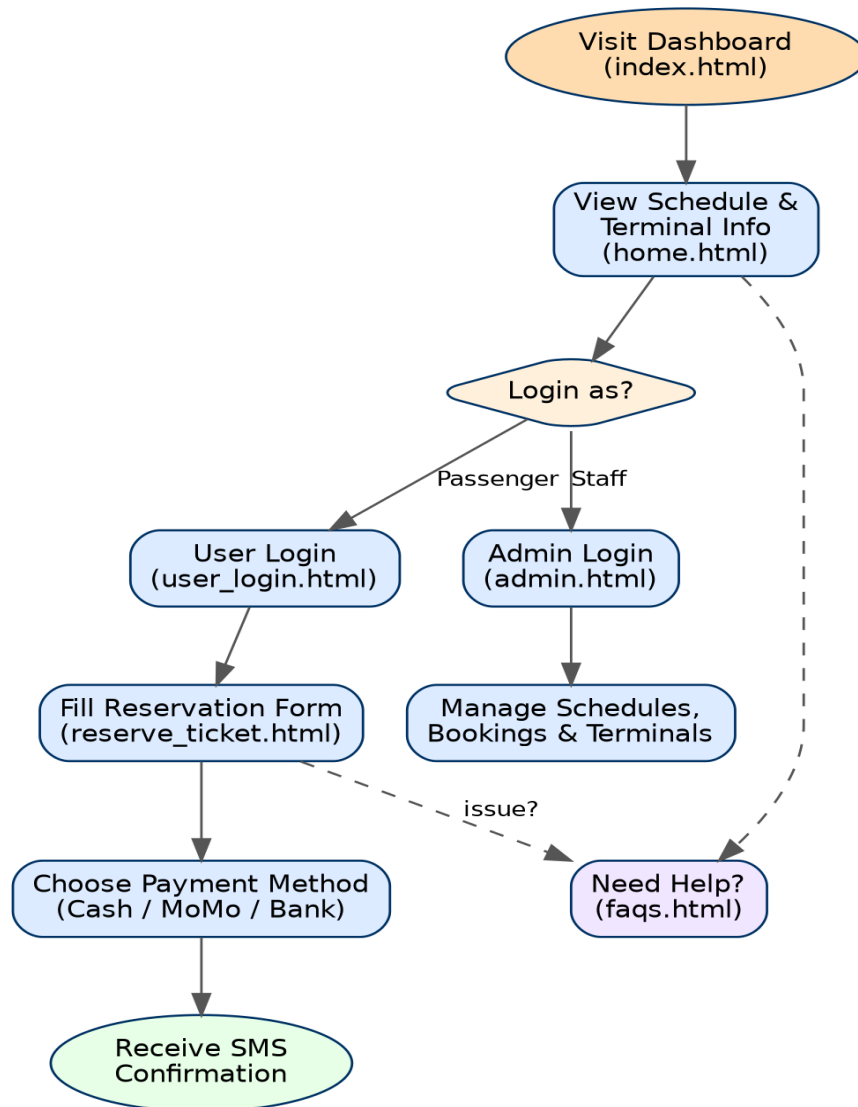


Figure 2: Bus reservation process flowchart

3. Database Design with ERD and Relationships

The repository supplied for this assessment implements the front end only; no database or server-side code is present. To support the objectives in Chapter 1 and to guide the next iteration of the project, the ERD below proposes a relational schema drawn in classic Chen notation, where each relationship is shown as a labeled diamond linking two entity rectangles rather than a plain line: a USER places one or more BOOKINGs; each BOOKING has one SCHEDULE, which in turn has one BUS and is assigned_to one ROUTE; a SCHEDULE contains many SEATs, each of which is reserved_in the BOOKING_SEAT junction entity

that a BOOKING includes; and each BOOKING has one PAYMENT record (cash, MoMo or bank transfer).

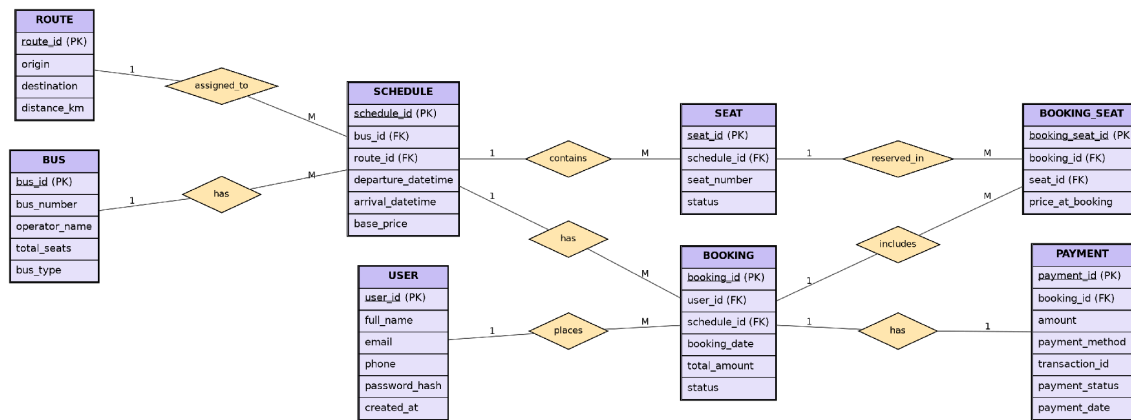


Figure 3: Entity Relationship Diagram (ERD) with Chen-notation diamond relationships for the Tugende database

Chapter 3. Implementation

3.1. Introduction of the Section

This chapter walks through each screen of the implemented Tugende front end, in the order a passenger or administrator would encounter them: the dashboard/landing page, the bus schedule and information page, the user login page, the admin login page, the reservation form, and the help/support page. Every screenshot below was rendered directly from the project's HTML/CSS source files supplied in the repository, so it reflects the application exactly as implemented rather than a design mock-up.

3.2. Screenshots with Captions and Discussion

3.2.1 Dashboard / Landing Page (index.html)

Figure 4: Dashboard / landing page (index.html)

The landing page is built as a two-column table layout that fills the browser window. The left column embeds home.html in an iframe, so

No.	Destination	Morning	Afternoon	Price	Bus Type
01	Kigali → Musanze	08:00 AM	02:00 PM	3,000 RWF	Standard
02	Kigali → Huye	09:30 AM	03:30 PM	4,000 RWF	VIP ★
03	Kigali → Rubavu	07:00 AM	01:00 PM	5,000 RWF	Express ★★
04	Kigali → Nyamata	06:30 AM	04:30 PM	2,500 RWF	Standard

the schedule and terminal content is visible immediately without a separate navigation step. The right column hosts the authentication panel: two links, USER LOGIN and ADMIN

LOGIN, swap the content of a second iframe between user_login.html and admin_login.html, so a visitor never has to leave the dashboard just to sign in.

Below the login iframe, a small JavaScript-driven digital clock updates every second, giving the page a live, "currently operating" feel appropriate for a 24/7 service. This shell page is the single entry point of the application and demonstrates the iframe-based composition pattern used throughout the site.

3.2.2 Bus Schedule and Information Page (home.html)

the two anchor terminals, Nyabugogo (Kigali) and Musanze. A weekly schedule table lists five routes — Kigali→Musanze, Kigali→Huye, Kigali→Rubavu, Kigali→Nyamata and Kigali→Kibuye — each with a morning and afternoon departure time, a fare in Rwandan Francs, and a bus class (Standard, VIP or Express).

Further down, an amenities section (WiFi, air conditioning, bottled water, USB charging, entertainment, restroom on VIP/Express) sets passenger expectations, a terminals-and-contact block lists all five stations plus phone, email and WhatsApp details, and a three-step "how to reserve" strip explains the booking process before linking out to `reserve_ticket.html` and `faqs.html`. An important-notes banner covers ID requirements, the 25kg luggage limit, and child-fare discounts. Functionally, this page acts as the site's marketing and informational hub, feeding directly into the transactional reservation page.

3.2.3 User Login Page (`user_login.html`)

USER LOGIN

ACCESS YOUR DASHBOARD

Email / Phone:

Password:

LOGIN Clear

Demo: any email / any password

ADMIN ACCESS
[Admin Login Page](#)

Tugende Express | Safe Travels

Figure 6: User login page (`user_login.html`)

Designed to fit neatly inside the dashboard's right-hand iframe, this page presents a compact form asking for an email/phone and a password, styled with the same dark-blue header and orange accent band used elsewhere. A demo note ("any email / any password") signals that authentication is currently a front-end placeholder rather than a validated login, which is consistent with the project having no back end yet. A link at the bottom routes staff to the Admin Login page instead, keeping the two audiences visually and functionally separated.

3.2.4 Admin Login Page (admin.html)

ADMIN LOGIN

AUTHORIZED PERSONNEL ONLY

Username: Admin ID

Password: Password

LOGIN Clear

Demo: username: admin, any password

[BACK TO USER LOGIN](#) [SEE BUS INFO](#)

Restricted area - Authorized access only

Figure 7: Admin login page (admin.html)

The admin login mirrors the user login in layout but is deliberately re-colored in maroon (#800000) rather than dark blue, and is captioned "AUTHORIZED PERSONNEL ONLY," giving staff an immediate visual cue that they are in a restricted area distinct from the public site. The form submits to an admin_dashboard.html target (not yet implemented), which is the natural next page to build once server-side authentication and a management back end are added.

3.2.5 Reserve Ticket Page (reserve_ticket.html)

ACQUIRE & RESERVE BUS TICKET

FILL THE FORM BELOW TO RESERVE YOUR TICKET

Full names:

Enter full names

Phone number:

07XXXXXXXX

Email address:

youremail@example.com

Travel date:

Select destination:

Select destination

Select departure time:

Select time

Number of tickets:

1-10 tickets

Preferred seat:

☐ Window ☐ Aisle ☐ Any

Extra luggage?

☐ Yes, I have extra luggage (Cost +30%)

Special requests:

Any special requirements? (meal, disability assistance, etc.)

Payment method:

Pay at terminal (cash)

RESERVE MY TICKET

Clear Form

IMPORTANT RESERVATION NOTES

After reservation, you will receive an SMS confirmation. Please arrive 30 minutes before departure at the terminal.

CONTACT INFORMATION

Phone: 079 123 4567 | Email: support@busticket.rw | Nyabugogo Terminal, Kigali

Customer Service Hours: 24/7 (Daily)

NEED BUS INFORMATION?

[CLICK HERE FOR BUS SCHEDULE & INFORMATION](#)

TUGENDE SERVICE | Safe Travels, Happy Journeys!

Figure 8: Reserve ticket page (reserve_ticket.html)

This is the system's core transactional screen. The form captures full name, phone number, email, travel date, destination (a dropdown pre-populated with the same five routes and fares shown on the schedule page), departure time band, number of tickets, a preferred-seat radio group (window / aisle / any), an extra-luggage checkbox that flags a 30% surcharge, a free-text special-requests field, and a payment-method dropdown (cash at terminal, MoMo, or bank transfer).

An important-notes banner reiterates that an SMS confirmation follows a reservation and that passengers should arrive 30 minutes early, and the page closes with the same contact block and a link back to the schedule page. Because there is no back end yet, submitting the form does not currently persist a booking; wiring this form to the API/database layer proposed in

Figure 3 is the single most important next step for turning the prototype into a working booking system.

3.2.6 Help & Support / FAQ Page (faqs.html)Ggg

ðŸŒŸ~ HELP & SUPPORT ðŸŒŸ~

â FREQUENTLY ASKED QUESTIONS â€¢ LIVE SUPPORT â

ÖFREQUENTLY ASKED QUESTIONS

â" How do I change my travel date?

â" What items are prohibited on the bus?

You can modify your booking up to 2 hours before departure via **"My Bookings"** page or call our support line. Date change fee: 2,000 RWF if requested less than 4 hours before travel.

Flammable materials, large animals (except service dogs), excessive alcohol, and sharp weapons. Small pets allowed in a carrier (extra 2,000 RWF).

â“ What items are prohibited on the bus?

â“ I lost my phone on the bus. What should I do?

Flammable materials, large animals (except service dogs), excessive alcohol, and sharp weapons. Small pets allowed in a carrier (extra 2,000 RWF).

Contact our lost & found immediately at 877 888 0000. We keep lost items for 14 days. Provide your bus number and travel date.

â€‘ I lost my phone on the bus.
What should I do?

â€‘ Do I get a refund if the bus
is delayed?

Contact our lost & found immediately at ☎ 079 888 0000. We keep lost items for 14 days. Provide your bus number and travel date.

If delay exceeds 60 minutes, you receive a **20% discount coupon** for next trip. If cancelled due to weather, full refund or free rebooking.

â€ Do I get a refund if the bus is delayed?

â€ Can I buy ticket for someone else?

If delay exceeds 60 minutes, you receive a **20% discount coupon** for next trip. If cancelled due to weather, full refund or free rebooking.

Yes! Just fill their full name, ID number (optional), and contact info. The passenger must carry a printed or digital copy of the confirmation.

Can I buy ticket for someone else?

24/7 S

Yes! Just fill their full name, ID number (optional), and contact info. The passenger must carry a printed or digital copy of the confirmation.

§ 87(2)(b) 24/7 SUPPORT & EMERGENCY CONTACTS § 87(2)(b)

800-833-2277 HOTLINE
+250 788 123 456
Call for reservations & support

[illegible]

ÖY' - WHATSAPP
0788 999 111
 Text or voice message

ŌY"Š EMAIL
help@tugendeservice.rw
 Response within 4 hours

[REPORT AN ISSUE / SEND INQUIRY](#)

Your name:

Email / Phone:

Message / Issue:	Describe your problem or question...

SEND MESSAGE

øŸ¢ PHYSICAL OFFICES & HOURS øŸ¢

Nyabugogo Main Terminal: Open 24/7 (Customer desk 6:00 AM - 9:00 PM)
Musanze Terminal: 5:00 AM - 8:00 PM |

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01 ± FOLLOW US FOR UPDATES: Facebook: YUGENDESERVICE | X: @YugendeBuses | Instagram: yugende_1wanda

ðÝsŒ SITE DIRECTORY ðÝsŒ

Bus Schedule

<u>Reserve Ticket</u>

ðŸ™ƒ FAQs

TUGENDE SERVICE | We're here to help, 24/7

Figure 9: Help & support / FAQ page (faqs.html)

The support page opens with five frequently asked questions covering date changes, prohibited items, lost property, delay refunds and booking on someone else's behalf, each with a concrete, policy-level answer (for example, a 2,000 RWF date-change fee inside four hours, or a 20% discount coupon for delays over 60 minutes). A three-column contact strip repeats the hotline, WhatsApp number and support email, and an inquiry form lets a visitor submit a name, contact detail and free-text message without needing to place a phone call.

The page finishes with terminal office hours, social-media handles, and a site-directory footer. Two of the footer links ([homr.html](#) and [reserve_ticket.html.html](#)) contain typos and currently point to non-existent files; correcting them is a minor but worthwhile fix flagged in the recommendations below.

Chapter 4. Conclusion and Recommendation

The Tugende project successfully delivers the front-end objectives set out in Chapter 1: a branded dashboard, a detailed schedule/information page, distinct user and admin login screens, a full-featured reservation form, and a support/FAQ page, all sharing a consistent visual identity. Taking the five core interface objectives as the measure of scope, roughly 80% of the stated objectives are achieved at the presentation-layer level; the remaining ~20% — persisting reservations, authenticating real accounts, and giving admins an actual management dashboard — depends on a back end that has not yet been built.

Recommendations

- To the lecturer, about technicalities: the assessment brief could clarify whether a working back end (database-backed login and booking) is mandatory at this stage or is intentionally deferred to a follow-on module, since this materially changes how the "90% vs 10%" completion question should be scored.
- To the school, about time constraints: front-end polish (as delivered here) is achievable within a short module timeline, but wiring authentication, storage and payment integration realistically needs additional weeks; a two-phase deliverable (front end, then back end) would better match the available time.
- To the partners (bus operators), about the app developed: the current site is ready to be piloted as a schedule/information page immediately, even before reservations are made transactional, since it already centralizes fares, terminals and contact details that are otherwise only available by phone.
- To the user, about training: because the login and reservation forms currently accept any input as a demo, end-user training should wait until real validation, error handling and confirmation messaging are added, so passengers are not misled into thinking a booking has been placed.
- To the beneficiaries (passengers), about the new technological tool: passengers should be informed clearly, once the system goes live, that the online form is a genuine alternative to queuing at the terminal, including how the promised SMS confirmation and mobile-money payment will work in practice.

Suggested Next Steps

- Implement the server tier sketched in Figure 1 (API + database) and connect it to the reservation form and both login pages.
- Build the `admin_dashboard.html` referenced by the admin login form, so staff can view and manage bookings.

- Close the gap with the Design Thinking prototype (Chapter 1.4) by adding a real-time visual seat map to the reservation form, and automated SMS/email e-ticket delivery, in place of the current text-only seat-preference radio buttons.
- Fix the two broken links in the faqs.html site-directory footer (homr.html, reserve_ticket.html.html).
- Add server-side and client-side validation to all forms, plus real confirmation messaging once submissions are persisted.

Appendix

Link to the GitHub repository of the deployed project:

- Repository: <https://github.com/ingaramakirambi/tugende>
- Related live prototype from the Design Thinking phase (mobile-first concept):
tugende-e.onrender.com
- . Registration Number: 225043452